

Physiology Practice Questions: DM Module

Lecture Titles: Exocrine Pancreas and Intestinal Transports (L26) and Clinical Cases of Biochemistry and Physiology (L39)

Note: If you need any clarification, please email to: ASK-PHY@squ.edu or email to the faculty who delivered this lecture.

1. A 41-year-old man comes to the physician for a follow-up examination. He has a 5-year history of chronic pancreatitis. He takes some enzyme supplements. His vital signs are within normal limits. Histological examination of a biopsy from the pancreas shows loss of exocrine pancreas due to chronic inflammation. Which of the following substances is most likely reduced in this patient's duodenal lumen?

- A. Pepsin
- B. Intrinsic factor
- C. Bile salts
- D. Bicarbonate
- E. Hydrochloric acid
- F. Secretin

2. A 24-year-old healthy woman participates in a study related to the fluid balance in the gastro-intestinal system, starting from mouth to colon. She is instructed to measure the fluid intake in a day. She consumes 2 liters of fluid in 24 hours in the form of food and juices. Which of the following statements best describes the relationship between fluid intake and gastro-intestinal secretions in this participant?

- A. The volume of fluid ingested equals that appears in the feces in a day
- B. Maximum amount of water absorption occurs in the stomach
- C. No water absorption takes place in the stomach
- D. All consumed fluid is typically absorbed in the colon
- E. The amount of fluid ingested is less than the amount absorbed in the entire GI in a day

3. Investigators conduct a study relating to the effect of various factors on gastrointestinal secretions. Investigators place acidic ($\text{pH} < 3$) fluid in the lumen of the duodenum and measure substances secreted from stomach, pancreas, and liver. Which of the following secretory alterations is most likely following this intervention?

- A. Serum gastrin increases
- B. Increases bicarbonate secretion from the pancreas
- C. Decreases bile production from the liver
- D. Increases gastric acid secretion
- E. Serum cholecystokinin increases

4. A 52-year-old woman comes to the emergency department because of a 25-minute history of upper abdominal pain. She has had 2 episodes of similar pain during the past 6 months. The pain started after she ate a fatty meal. Her BMI is 32 kg/m² (Normal: 19-25). An abdominal radiograph shows the presence of radiopaque structures in the right upper quadrant. She is diagnosed with gallstone disease. Which of the following additional findings is most likely during pain in this patient?

- A. Increased pepsinogen secretion from stomach
- B. Contraction of the stomach
- C. Increased acid production from the stomach
- D. Increased enzyme secretion by the pancreas
- E. Relaxation of the gallbladder

5. Investigators conduct a study relating to the effect of various endogenous agents on the exocrine pancreatic secretion. Investigators intravenously inject these substances either singly or in combinations and collect pancreatic juice from the main pancreatic duct. Composition of this juice is analyzed and recorded. Which of the following combinations of agents most likely produces the highest rate of pancreatic bicarbonate secretion?

- A. Secretin and histamine
- B. Cholecystokinin and acetylcholine
- C. Gastrin and acetylcholine
- D. Secretin and cholecystokinin
- E. Cholecystokinin and acetylcholine

6. A 32-year-old man is brought to the emergency department because of a 2-day history of profuse watery diarrhea and weakness. He recently returned from a trip to south Africa. Physical examination shows sunken eyes, dry mouth, and reduced skin turgor. Stool studies is positive for *Vibrio cholerae*. Which of the following mechanisms is the most likely cause of diarrhea in this patient?

- A. Activation of the bicarbonate channels in the intestinal epithelium cells
- B. Activation of chloride channels in the intestinal epithelial cells
- C. Inhibition of absorption of glucose in the intestine
- D. Inhibition of cAMP production in the intestinal epithelium
- E. Activation of ENaC channels in the intestinal epithelium

7. A 47-year-old man comes to the physician because of a 1-week history of diarrhea and weakness. He has a 3-year history of ulcerative colitis. Physical examination shows mild dehydration. Stool studies show increased volume of fluid that is secreted from the large intestine. Which of the following electrolyte loss in the stools best explains this patient's weakness?

- A. Sodium
- B. Chloride
- C. Bicarbonate
- D. Calcium
- E. Potassium

8. A 43-year-old man is brought to the emergency department because of a 3-hour history of severe abdominal pain associated with nausea and vomiting. The pain radiates his back. These symptoms began after a weekend of heavy drinking. He drinks 250 mL of vodka daily and a 12-pack of beer most weekends. His temperature is 38°C (99.8°F), pulse is 110/min, respirations are 20/min, and blood pressure is 128/84 mm Hg. Physical examination shows epigastric tenderness and bluish discoloration around the umbilicus. He is diagnosed with acute pancreatitis. In this condition, damage to the pancreas releases its enzymes into blood. Serum elevation in which of the following enzymes is most likely in this patient?

- A. Sucrase
- B. Pepsin
- C. Maltase
- D. Lipase
- E. Enteropeptidases

9. Investigators conduct a study related to the fluid movements in different parts of the gastrointestinal tract. For this purpose, they measure the amount of fluid that is ingested, fluid entering each part of the GIT, and secreted from each part of the GIT. Which of the following statements is true regarding fluid dynamics in the gastrointestinal tract in a normal healthy individual in a day?

- A. Only about 100 to 200 mL is excreted in the stools
- B. Most fluid in small intestine comes from diet
- C. Highest amount of fluid is absorbed in the large intestine
- D. Gastric secretions contribute twice the volume of those from the pancreas
- E. Most of the intestinal fluid comes from liver as bile

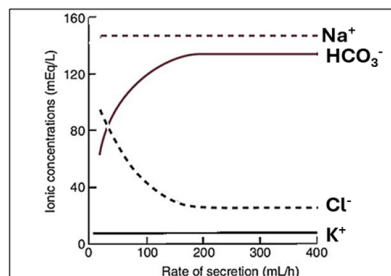
10. A 33-year-old man comes to the physician because of a 3-day history of watery diarrhea. He says he recently eat street foods several times during his vacation with family. Physical examination shows mild dehydration. Serum sodium and potassium concentrations are within normal reference ranges. Stool studies show E. coli and increased concentration of bicarbonate but not potassium. He is diagnosed with traveler's diarrhea because of E. coli infection. Which of the following acid-base imbalance is most likely in this patient?

- A. Metabolic alkalosis
- B. Metabolic acidosis
- C. Respiratory acidosis
- D. Respiratory alkalosis
- E. Mixed acidosis

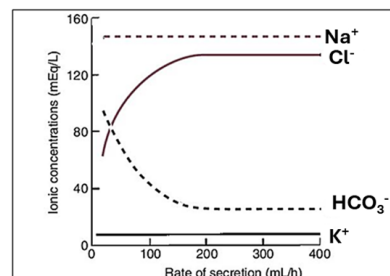
11. A 55-year-old man comes to the physician for a follow-up examination. He says he passes pale, foul-smelling stools. He was diagnosed with viral hepatitis (liver injury due to viral infection) six months ago. Physical examination shows jaundice and a palpable liver. Laboratory studies show elevated serum bilirubin and liver enzymes. His pancreas function is normal. Amount of bile in the duodenum is found to be reduced. Which of the following defects best explains defective fat digestion and absorption in this patient?

- A. Defective formation of micelles in the small intestine
- B. Decreased amount of pancreatic lipase in the small intestine
- C. Defective chylomicron formation in the intestinal epithelium
- D. Inability of the gallbladder contraction
- E. Deficiency of co-lipase in the small intestine

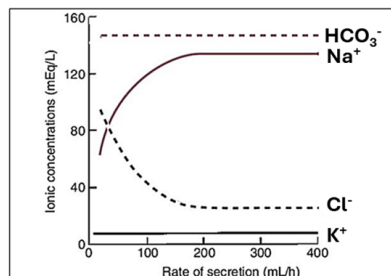
12. Investigators conduct a study related to the effect of rate of pancreatic secretion on its electrolyte composition in 20 healthy men of 25-year-old. Investigators collect pancreatic secretion by using various stimulants and record the electrolyte composition as shown in the graphs. Which of the following graphs best represents the normal response?



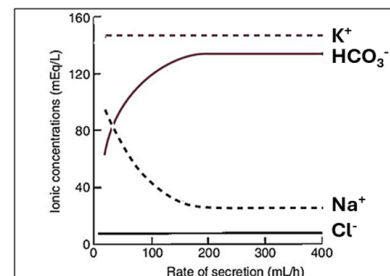
A



B



C



D

13. A 41-year-old man comes to the physician for a follow-up examination. He has had terminal ileal resection 8 months ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following gastrointestinal functional alterations is most likely in this patient?

- A. Increased concentration of primary bile acids in bile
- B. Decreased concentration of bile in the gall bladder
- C. Increases the rate of biliary duct secretion of bicarbonate
- D. Increases dietary lipid absorption
- E. Increases fecal bile acid excretion

14. A 3-year-old girl is brought to the emergency department by her mother because of a 2-day history of vomiting, diarrhea, and lethargy. The mother says she has fever for 3 days before these symptoms appeared and she gave her Panadol (fever lowering drug). Her temperature is 38°C (100.4°F), pulse is 154/min, and blood pressure is 90/54 mm Hg. Physical examination shows dry mouth, tongue, low skin turgor, and increased capillary refill time. She is diagnosed with acute viral gastroenteritis. She is resuscitated with intravenous fluids. After 6 hours of this intervention, she is started with oral fluids. She is prescribed an oral dehydration solution that contains glucose, sodium, and water. Which of the following best explains the need for adding glucose in the solution prescribed to this patient?

- A. Glucose is readily metabolized to provide energy
- B. Glucose enhances absorption of water in the intestine
- C. Needed as substrate for the cotransporter to absorb sodium in the intestine
- D. Increases glycogen store in the liver
- E. Stimulates gut hormones to increase digestion and absorption in the intestine

15. Investigators study the effect of hormones on pancreatic secretion during different phases of digestion. 50 healthy volunteers of both gender are selected for this study. These participants are provided with standard diet. The volume of pancreatic secretion, serum hormone concentrations are measured during each phase of digestion, and the composition of pancreatic secretions are analyzed in the laboratory. Which of the following sets of alterations is most likely regarding pancreatic secretion in these participants?

Option	Volume of pancreatic secretion	Phase of digestion	Major hormones elevated
A	Minimal	Cephalic phase	Gastrin and secretin
B	Maximal	Gastric phase	CCK and secretin
C	Moderate	Intestinal phase	CCK and gastrin
D	Maximal	Intestinal phase	CCK and secretin
E	Moderate	Gastric phase	Gastrin and VIP

Answer Key and Explanation

1. A 41-year-old man comes to the physician for a follow-up examination. He has a 5-year history of chronic pancreatitis. He takes some enzyme supplements. His vital signs are within normal limits. Histological examination of a biopsy from the pancreas shows loss of exocrine pancreas due to chronic inflammation. Which of the following substances is most likely reduced in this patient's duodenal lumen?

- A. Pepsin
- B. Intrinsic factor
- C. Bile salts
- D. Bicarbonate
- E. Hydrochloric acid
- F. Secretin

Answer Key: D

Rationale: This patient has damaged exocrine pancreas which secretes digestive enzymes and bicarbonate. Both these constituents are deficient in the duodenum.

2. A 24-year-old healthy woman participates in a study related to the fluid balance in the gastro-intestinal system, starting from mouth to colon. She is instructed to measure the fluid intake in a day. She consumes 2 liters of fluid in 24 hours in the form of food and juices. Which of the following statements best describes the relationship between fluid intake and gastro-intestinal secretions in this participant?

- A. The volume of fluid ingested equals that appears in the feces in a day
- B. Maximum amount of water absorption occurs in the stomach
- C. No water absorption takes place in the stomach
- D. All consumed fluid is typically absorbed in the colon
- E. The amount of fluid ingested is less than the amount absorbed in the entire GI in a day

Answer Key: E

Rationale: The total daily secretion of fluid starting from saliva until colon is about 7 Liters. She consumes just 2 liters per day. Total fluids getting into GI is about 9 Liters. Of this, about 98% is reabsorbed in different parts of the GI (from stomach until colon) and feces contains about 100 mL of water. Maximum water absorption occurs in the small intestine.

3. Investigators conduct a study relating to the effect of various factors on gastrointestinal secretions. Investigators place acidic (pH<3) fluid in the lumen of the duodenum and measure substances secreted from stomach, pancreas, and liver. Which of the following secretory alterations is most likely following this intervention?

- A. Serum gastrin increases

- B. Increases bicarbonate secretion from the pancreas
- C. Decreases bile production from the liver
- D. Increases gastric acid secretion
- E. Serum cholecystokinin increases

Answer Key: B

Rationale: Acidic chyme entering the duodenum causes the release of secretin from S cells of the duodenum. Secretin is a hormone that acts on the pancreas to increase bicarbonate secretion from ductal cells. The acid-induced feedback from the duodenum cause the opposite effects in options A and D and has no direct effect on bile production.

4. A 52-year-old woman comes to the emergency department because of a 25-minute history of upper abdominal pain. She has had 2 episodes of similar pain during the past 6 months. The pain started after she ate a fatty meal. Her BMI is 32 kg/m² (Normal: 19-25). An abdominal radiograph shows the presence of radiopaque structures in the right upper quadrant. She is diagnosed with gallstone disease. Which of the following additional findings is most likely during pain in this patient?

- A. Increased pepsinogen secretion from stomach
- B. Contraction of the stomach
- C. Increased acid production from the stomach
- D. Increased enzyme secretion by the pancreas
- E. Relaxation of the gallbladder

Answer Key: D

Rationale: Fatty meal entering the duodenum causes the release of CCK from I cells. CCK acts on the pancreas to increase enzyme-rich secretion, from the acinar cells. CCK stimulates smooth muscle contraction in the gall bladder but relaxes the smooth muscles in stomach and sphincter of Oddi.

5. Investigators conduct a study relating to the effect of various endogenous agents on the exocrine pancreatic secretion. Investigators intravenously inject these substances either singly or in combinations and collect pancreatic juice from the main pancreatic duct. Composition of this juice is analyzed and recorded. Which of the following combinations of agents most likely produces the highest rate of pancreatic bicarbonate secretion?

- A. Secretin and histamine
- B. Cholecystokinin and acetylcholine
- C. Gastrin and acetylcholine
- D. Secretin and cholecystokinin
- E. Cholecystokinin and acetylcholine

Answer Key: D

Rationale: Secretin is the key stimulus for pancreatic bicarbonate secretion, so we can narrow it down to answers A and D. Histamine acts on parietal but not on pancreatic ductal cells, whereas CCK also acts on the pancreatic ducts to further stimulate bicarbonate secretion, hence D is the best answer.

6. A 32-year-old man is brought to the emergency department because of a 2-day history of profuse watery diarrhea and weakness. He recently returned from a trip to south Africa. Physical examination shows sunken eyes, dry mouth, and reduced skin turgor. Stool studies is positive for *Vibrio cholerae*. Which of the following mechanisms is the most likely cause of diarrhea in this patient?

- A. Activation of the bicarbonate channels in the intestinal epithelium cells
- B. Activation of chloride channels in the intestinal epithelial cells
- C. Inhibition of absorption of glucose in the intestine
- D. Inhibition of cAMP production in the intestinal epithelium
- E. Activation of ENaC channels in the intestinal epithelium

Answer Key: B

Rationale: *Vibrio cholerae* is a bacteria that produces profuse watery (secretory) diarrhea. This organism produces a toxin called cholera toxin that stimulates chloride channels in the intestinal epithelium. This effect is mediated via cAMP. Chloride leakage into the lumen makes movement of sodium (electrical movement) and water (osmotic movement) into the lumen resulting in secretory diarrhea.

7. A 47-year-old man comes to the physician because of a 1-week history of diarrhea and weakness. He has a 3-year history of ulcerative colitis. Physical examination shows mild dehydration. Stool studies show increased volume of fluid that is secreted from the large intestine. Which of the following electrolyte loss in the stools best explains this patient's weakness?

- A. Sodium
- B. Chloride
- C. Bicarbonate
- D. Calcium
- E. Potassium

Answer Key: E

Rationale: There is net secretion of potassium in the large intestine. This patient's diarrhea is mostly due to secretion of large intestinal fluid which contains high amount of potassium. Increased potassium loss (hypokalemia) causes muscle weakness because of its effect on resting membrane potential. Hypokalemia induces metabolic alkalosis. If a patient loses large amount of small intestinal fluid, the major electrolyte lost is bicarbonate. If fecal bicarbonate loss becomes predominant, the patient will have metabolic acidosis.

8. A 43-year-old man is brought to the emergency department because of a 3-hour history of severe abdominal pain associated with nausea and vomiting. The pain radiates his back. These symptoms began after a weekend of heavy drinking. He drinks 250 mL of vodka daily and a 12-pack of beer most weekends. His temperature is 38°C (99.8°F), pulse is 110/min, respirations are 20/min, and blood pressure is 128/84 mm Hg. Physical examination shows epigastric tenderness and bluish discoloration around the umbilicus. He is diagnosed with acute pancreatitis. In this condition, damage to the pancreas releases its enzymes into blood. Serum elevation in which of the following enzymes is most likely in this patient?

- A. Sucrase
- B. Pepsin
- C. Maltase
- D. Lipase
- E. Enteropeptidases

Answer Key: D

Rationale: Pancreas secretes major digestive enzymes that includes lipase, amylase, trypsinogen, chymotrypsinogen, elastase, carboxypeptidase, and cholesterol esterase. Damage to the pancreas causes release of some of these enzymes into blood and raises its concentration. Elevated serum concentrations of amylase and lipase are the important laboratory findings in diagnosing pancreatitis. Other options (A,B,C,E) are not secreted from pancreas.

9. Investigators conduct a study related to the fluid movements in different parts of the gastrointestinal tract. For this purpose, they measure the amount of fluid that is ingested, fluid entering each part of the GIT, and secreted from each part of the GIT. Which of the following statements is true regarding fluid dynamics in the gastrointestinal tract in a normal healthy individual in a day?

- A. Only about 100 to 200 mL is excreted in the stools
- B. Most fluid in small intestine comes from diet
- C. Highest amount of fluid is absorbed in the large intestine
- D. Gastric secretions contribute twice the volume of those from the pancreas
- E. Most of the intestinal fluid comes from liver as bile

Answer Key: A

Rationale: Most of the fluid entering the GI tract is re-absorbed (about 98%) and only about 100-200 mL/day are excreted in feces. Most of this fluid in the small intestine comes from GI secretions like gastric, bile, pancreatic and not from diet. Highest amount of fluid is absorbed in the small intestine but not in the large intestine. Pancreatic and gastric secretions are roughly the same volume per day, about 1-2 Liters. Volume of bile entering

the small intestine is about 500 mL/day and is not the highest contributor in the small intestine.

10. A 33-year-old man comes to the physician because of a 3-day history of watery diarrhea. He says he recently eat street foods several times during his vacation with family. Physical examination shows mild dehydration. Serum sodium and potassium concentrations are within normal reference ranges. Stool studies show *E. coli* and increased concentration of bicarbonate but not potassium. He is diagnosed with traveler's diarrhea because of *E. coli* infection. Which of the following acid-base imbalance is most likely in this patient?

- A. Metabolic alkalosis
- B. Metabolic acidosis
- C. Respiratory acidosis
- D. Respiratory alkalosis
- E. Mixed acidosis

Answer Key: B

Rationale: He has lost large amount of bicarbonate because his stool contains increased amount of bicarbonate. This results in metabolic acidosis. It is also possible to have metabolic alkalosis due to chronic diarrhea. If a patient loses large amount of potassium in stools, it results in hypokalemia. Hypokalemia induces metabolic alkalosis. Therefore, both types of metabolic acid-base disturbances are possible in a patient with diarrhea. It depends on what is predominantly (bicarbonate or potassium) lost in feces.

11. A 55-year-old man comes to the physician for a follow-up examination. He says he passes pale, foul-smelling stools. He was diagnosed with viral hepatitis (liver injury due to viral infection) six months ago. Physical examination shows jaundice and a palpable liver. Laboratory studies show elevated serum bilirubin and liver enzymes. His pancreas function is normal. Amount of bile in the duodenum is found to be reduced. Which of the following defects best explains defective fat digestion and absorption in this patient?

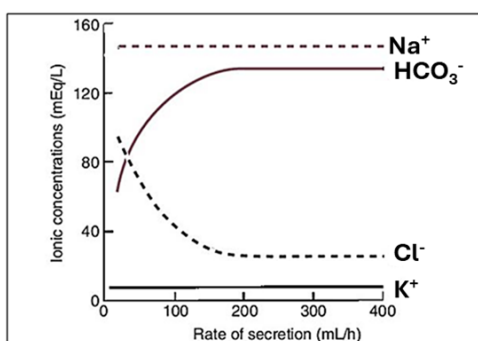
- A. Defective formation of micelles in the small intestine
- B. Decreased amount of pancreatic lipase in the small intestine
- C. Defective chylomicron formation in the intestinal epithelium
- D. Inability of the gallbladder contraction
- E. Deficiency of co-lipase in the small intestine

Answer Key: A

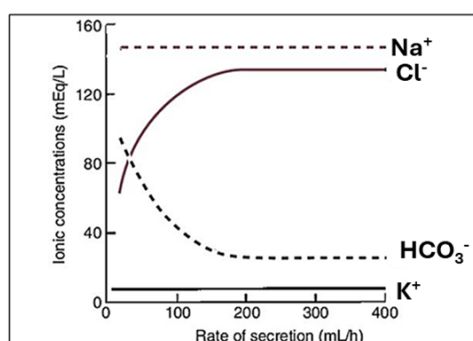
Rationale: Bile acids/bile salts are essential for fat digestion and absorption in the intestine. They have 2 functions: Emulsification and micelle formation. Emulsification convert dietary fat into small lipid droplets that increase the surface area and facilitates enzymes to digest fat. Digestive enzymes coming from the pancreas act on the lipid droplets and split them into end products of digestion of fat namely fatty acids,

monoacylglycerol (MAG), lysolecithins, and lipid-soluble vitamins. Next stage is micelle formation. These are tiny structures that has bile salts on the surface and end products of fat digestion in the core. These micelles become water-soluble and can then be absorbed through the intestinal epithelium. If bile salts are absent (or deficient), as in the above patient, the micelle formation fails which results in reduced fat absorption.

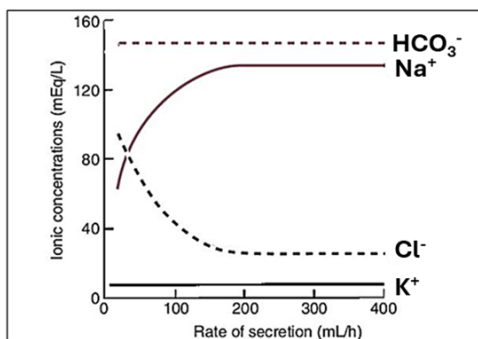
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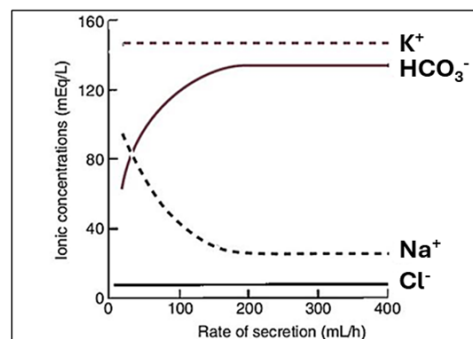
A



B



C



D

Answer Key: A

Rationale: When pancreas is not stimulated by hormone, secretin, the rate of secretion is low. This is called basal (low) rate during which the pancreatic cells secrete an isotonic solution composed mainly of Na^+ , Cl^- , and water. When pancreas is stimulated by secretin, the rate of secretion increases. Secretin stimulates the ductal cells to secrete bicarbonate in exchange with chloride through a $\text{HCO}_3^-/\text{Cl}^-$ exchanger. This results in a greater concentration of bicarbonate and reduced concentration of chloride during the higher rate of secretion. The sodium and potassium concentrations in the pancreatic juice remain the same irrespective of the rate of secretion.

13. A 41-year-old man comes to the physician for a follow-up examination. He has had terminal ileal resection 8 months ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following gastrointestinal functional alterations is most likely in this patient?

- A. Increased concentration of primary bile acids in bile
- B. Decreased concentration of bile in the gall bladder
- C. Increases the rate of biliary duct secretion of bicarbonate
- D. Increases dietary lipid absorption
- E. Increases fecal bile acid excretion

Answer Key: E

Rationale: The terminal ileum is the key site of bile acid/bile salt absorption in the GIT. If that part is removed (ileal resection), bile salts cannot be absorbed anymore (enterohepatic circulation of bile salt reduces). This results in greater amount of bile salts entering the colon and are lost in feces. In this condition, liver is stimulated to synthesize more amount of fresh primary bile acids. Less concentration of bile salts in bile reduces fat digestion and absorption. Another equally important substance that is absorbed in the terminal ileum is vitamin B12.

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- A. Glucose is readily metabolized to provide energy
- B. Glucose enhances absorption of water in the intestine
- C. Needed as substrate for the cotransporter to absorb sodium in the intestine
- D. Increases glycogen store in the liver
- E. Stimulates gut hormones to increase digestion and absorption in the intestine

Answer Key: C

Rationale: During diarrhea, large volume of fluid and sodium is lost from the body. This must be replenished by oral administration of sodium chloride solution. The absorption of sodium in the intestine requires glucose as both substances are transported together by a Sodium-glucose co-transporter (SGLT type1). Adding glucose to the oral rehydration solution facilitates intestinal sodium absorption.

15. Investigators study the effect of hormones on pancreatic secretion during different phases of digestion. 50 healthy volunteers of both gender are selected for this study. These participants are provided with standard diet. The volume of pancreatic secretion, serum hormone concentrations are measured during each phase of digestion, and the composition of pancreatic secretions are analyzed in the laboratory. Which of the following sets of alterations is most likely regarding pancreatic secretion in these participants?

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C	Moderate	Intestinal phase	CCK and gastrin
D	Maximal	Intestinal phase	CCK and secretin
E	Moderate	Gastric phase	Gastrin and VIP

Answer Key: D

Rationale: Pancreatic secretion, like gastric secretion, occurs in all three digestive phases namely, cephalic, gastric, and intestinal phase. The intestinal phase is the most important phase which begins when chyme enters the duodenum (small intestine). Chyme stimulates secretion of CCK and secretin that increases the pancreatic secretion rich in enzymes and bicarbonate. The volume of pancreatic secretion is maximal during this phase.